VOLUME III

AI Architecture

Purpose

Volume III specifies the artificial intelligence architecture of Project APEX. It describes how observations are transformed into memory, reasoning, predictions, plans, and executable actions using modular AI components.

Vision

The AI architecture is designed to evolve from passive observation to explainable assistance and controlled autonomy while preserving user oversight, privacy, and adaptability.

Volume III Structure

Chapter 14 — LLM Layer

Natural-language reasoning, planning, and orchestration.

Chapter 15 — Vision Layer

Screen understanding, UI perception, and multimodal inputs.

Chapter 16 — Memory Layer

Working, episodic, semantic, and procedural memory.

Chapter 17 — Retrieval-Augmented Generation (RAG)

Grounding reasoning with workflow memory and documents.

Chapter 18 — Knowledge Graph

Relationships among users, workflows, skills, tools, and context.

Chapter 19 — Behavior Transformer

Learning behavioral patterns from historical workflows.

Chapter 20 — Agent Loop

Observe → Think → Plan → Execute → Reflect cycle.

Chapter 21 — Human Feedback Learning

Continuous improvement using user corrections and approvals.

Chapter 22 — Multi-Agent Architecture

Specialized collaborating agents for observation, planning, execution, and evaluation.

Expected Outcomes

- Unified AI architecture for Project APEX
- Explainable reasoning pipeline
- Behavior-aware planning
- Reliable long-term memory
- Context-aware autonomous execution
- Scalable multi-agent framework